

Responsible AI – Financial fraud – the brief

Team: Brian Tarran, Matthew Agius, Dharm Sivarajah

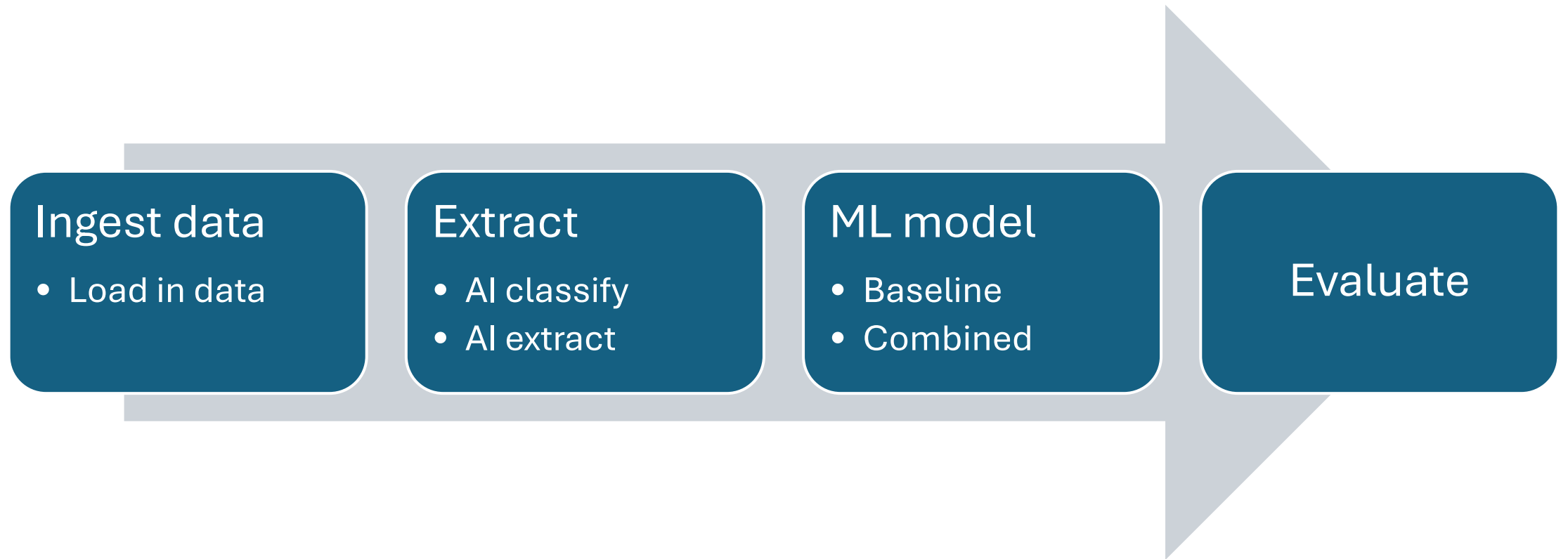
Context

Financial fraud is relevant to HMRC, DWP, FCA, and the Home Office. Traditional rule-based and statistical fraud detectors operate on structured fields — amounts, frequencies, counterparties. But fraud signals are also present in unstructured text: transaction narratives, company filings, communications. LLM-derived semantic features can complement and enrich structured models, but the stakes are high — a false positive has real consequences for the person or business flagged.

Goal

Combine structured financial features with LLM-extracted semantic features from text to improve anomaly detection or fraud classification. Demonstrate that the combined feature set outperforms structured features alone, and that the model's reasoning is interpretable by a human investigator.

Overall process chart



Work process (detailed)

1. Ingest data: Read each narrative using AI.
 - Ensuring no protected attributes used (**P2**)
2. Extract data:
 - Classify the fraud type compare against 5 definitions and decide best fitting one.
 - Pump and Dump
 - Rug Pull
 - Wash Trading
 - Coordinated Accumulation
 - Liquidity Manipulation
 - Extract useful information e.g.
 - Tactic
 - Entities
 - Supply Concentration
 - Urgency
 - Risk
 - Confidence
3. Converts AI output to columns and prints a summary for investigators
 - Combines columns for investigator review (**P4**): scheme, tactic, risk, urgency, supply concentration
4. Train baseline model
 - Convert fraud column to 1,0
 - Split data into train and test (70/30)
 - Baseline model using only price numeric column
 - Predicts fraud probability
5. Combined model
 - Model now sees price change, fraud scheme, risk, urgent, confidence, supply concentration
 - Train combined model
 - Combined prediction
6. Performance metrics
 - Precision
 - Recall
 - F1
7. Calculate Uplift

Performance metrics

=====						=====					
BASELINE - raw numeric signals only						COMBINED - numeric + LLM semantic features					
=====						=====					
ROC-AUC	:	0.9898				ROC-AUC	:	0.9974			
PR-AUC	:	0.9932				PR-AUC	:	0.9984			
Precision	:	0.9940				Precision	:	0.9832			
Recall	:	0.9222				Recall	:	0.9778			
F1	:	0.9568				F1	:	0.9805			
		precision	recall	f1-score	support			precision	recall	f1-score	support
	0	0.89	0.99	0.94	120		0	0.97	0.97	0.97	120
	1	0.99	0.92	0.96	180		1	0.98	0.98	0.98	180
accuracy				0.95	300	accuracy				0.98	300
macro avg		0.94	0.96	0.95	300	macro avg		0.98	0.98	0.98	300
weighted avg		0.95	0.95	0.95	300	weighted avg		0.98	0.98	0.98	300
=====						=====					
UPLIFT FROM LLM FEATURES											
=====											
ROC-AUC	delta :	+0.0080	(0.9898	→	0.9978)						
PR-AUC	delta :	+0.0054	(0.9932	→	0.9987)						

Principle 10: Model Card

Purpose

Improve fraud classification using LLM-extracted semantic features from text

Inputs

Synthetic transaction narratives – pre-generated text descriptions for LLM feature extraction

Outputs

Precision tables displaying model accuracy

Performance

99.74% ROC-AUC score, compared to a 98.98% baseline